

Perfect Retrieval Recall on ConvoMem:

MemoryKG’s Hybrid Semantic-Graph Architecture
Achieves 100% Across 17,463 Items

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Abstract

We evaluate MemoryKG, a conversational memory system based on DocKG [Suchanek, 2026b], on the ConvoMem benchmark [Pakhomov et al., 2025], a dataset of 75,336 QA pairs spanning six evidence categories that require locating one to six specific messages within a multi-session conversation corpus. MemoryKG achieves **100% retrieval recall** across all categories and evidence tiers (1–4 messages), tested over **17,463 items** at up to 1,000 items per category per tier — the largest evaluation on ConvoMem reported by any non-LLM retrieval system. MemoryKG outperforms MemPal (92.9% overall on a 300-item sample), the closest comparable verbatim-storage baseline, with the largest gains on the hardest categories: Preferences (+14.0 pp), Implicit Connections (+10.7 pp), and Abstention (+9.0 pp). The primary architectural advantage over flat vector search is *graph expansion*: semantic seeds are propagated through structured graph edges (CONTAINS, NEXT, REFERENCES), recovering evidence that embedding similarity alone misses when question and answer use different vocabulary. A single shared sentence-transformer embedder (BAAI/bge-small-en-v1.5) processes each per-item corpus in approximately 0.07 seconds on Apple Silicon.

1 Introduction

Conversational memory retrieval — the ability to locate a specific past message given a natural-language question — is a core capability for long-horizon AI assistants. Prior approaches range from full-context LLM inference [Pakhomov et al., 2025], through LLM-extracted structured memory [Mem0, 2024], to flat dense retrieval over verbatim session text [MemPal, 2026]. Each trades retrieval accuracy, latency, cost, and privacy differently.

The ConvoMem benchmark [Pakhomov et al., 2025] provides a systematic evaluation across six evidence categories, each requiring the system to locate between one and six specific “evidence messages” within a conversation corpus of up to 300 sessions. The benchmark paper finds that long-context LLMs (Gemini 2.5 Pro) score 89.4% on user-fact questions at 300-conversation scale, while RAG-based systems like Mem0 degrade from 61% (1-message evidence) to 25% (6-message evidence).

MemoryKG [Suchanek, 2026a] takes a different approach: it builds a lightweight knowledge graph over verbatim conversation text, where semantic vector search seeds initial candidates and structured graph traversal expands the result set. MemoryKG is a direct specialisation of DocKG [Suchanek, 2026b], a hybrid semantic-graph system for general document corpora. MemoryKG adapts the DocKG graph architecture specifically for multi-session conversational memory, replacing DocKG’s document-oriented corpus ingestion with per-session Markdown serialisation. No LLM inference is required at any stage — neither for indexing nor for retrieval. In this paper we report MemoryKG’s performance on ConvoMem and analyse why the graph expansion step closes the gap between flat vector search and perfect recall.

2 The ConvoMem Benchmark

ConvoMem [Pakhomov et al., 2025] contains 75,336 QA pairs drawn from multi-session dialogues. Each item consists of a conversation corpus, a question, and a set of *evidence messages* — the specific turns whose content is needed to answer the question correctly. Items are stratified by the number of evidence messages required (“tier”): tier 1 requires one message, tier 2 requires two, and so on up to tier 6.

The six evidence categories are:

User Facts Factual claims the user stated about themselves.

Assistant Facts Claims the assistant made during the conversation.

Abstention Cases where the assistant should decline to answer due to insufficient evidence.

Implicit Connections Answers requiring connecting information across multiple conversational contexts.

Preferences User preferences expressed directly or indirectly.

Changing Facts Facts that evolved over the course of the conversation; by construction, changing evidence begins at tier 2.

We use *retrieval recall* as our metric: the fraction of evidence messages whose text is found verbatim in the top- k retrieved nodes. In plain terms: for each question, we ask whether the system returned the specific conversation turns that contain the answer. A recall of 1.000 means every required message was found every time; a recall of 0.929 means roughly 1 in 14 questions returned an incomplete set. This is a binary retrieval metric, not end-to-end QA accuracy. The distinction matters: the ConvoMem paper reports QA accuracy (which requires an LLM to generate an answer), whereas we report whether the evidence is present in the retrieved context. These are different but complementary measures of system capability.

3 MemoryKG Architecture

MemoryKG represents a conversation corpus as a four-layer knowledge graph:

1. **Document layer.** Each conversation is a root document node containing the full session text.
2. **Section layer.** Markdown headings (one per speaker turn) produce section nodes connected to their parent document by CONTAINS edges.
3. **Chunk layer.** Section content is split into sentence-aligned chunks; adjacent chunks are linked by NEXT edges.
4. **Semantic index.** All nodes are embedded with a sentence transformer and stored in LanceDB for approximate nearest-neighbour retrieval.

3.1 Query pipeline

A query q is processed in two stages:

Stage 1 — Semantic seeding. The top- k nodes by cosine similarity to the query embedding are retrieved from LanceDB. These form the *seed set* S .

Stage 2 — Graph expansion. Starting from S , BFS expansion walks outward through the graph for h hops along any edge in the relation set $\mathcal{R}$. The default relation set is $\mathcal{R} = \{\text{CONTAINS}, \text{NEXT}, \text{REFERENCES}, \text{SIMILAR_TO}, \text{HAS_TOPIC}, \text{MENTIONS_ENTITY}, \text{HAS_KEYWORD}\}$. Nodes reachable within h hops of any seed are added to the candidate pool and ranked by a composite key combining seed distance, hop depth, and structural position.

For ConvoMem evaluations we use $k=10$ seeds and $h=1$ hop.

3.2 Why graph expansion helps

Consider a Preferences question: “What kind of coffee does the user prefer?” The evidence message might read “I’ve always found dark roasts a bit much for me.” The semantic similarity between the question and the evidence is moderate at best — the vocabulary is disjoint. A flat vector search may rank this message outside the top 10. Graph expansion rescues it: if any adjacent chunk (e.g. “Do you want some coffee? I usually go for a medium roast”) ranks in the seed set, the NEXT edge walks to the evidence chunk and adds it to the candidate pool. This structural propagation is the mechanism behind MemoryKG’s gains on Preferences (+14.0 pp over MemPal) and Implicit Connections (+10.7 pp).

4 Experimental Setup

4.1 Corpus format

For each evidence item, all conversations are written to a temporary directory as Markdown files (one file per conversation). Each speaker turn becomes a ##-headed section:

```
# Conversation 0

## User

Hey, I've been trying to optimize my lead tracking...

## Assistant

Sure! Do you use color-coding?

## User

Yeah --- I use green for hot leads in my personal spreadsheet.
```

The heading chunk strategy splits at ## boundaries, so each turn becomes one chunk node.

4.2 Build configuration

- Chunk strategy: **heading** (no per-file embedding at parse time)
- Topic / entity / keyword enrichment: disabled (speed)
- SIMILAR_TO edge discovery: disabled (speed)
- Workers: 1 (per-item builds are already fast)
- Batch size: 512 embeddings

4.3 Shared embedder

A single `SentenceTransformerEmbedder` (BAAI/bge-small-en-v1.5, 384-dimensional) is initialised once and reused across all items. This avoids the per-item model reload overhead and is the primary factor behind the 0.07 s/item throughput.

4.4 Recall computation

For each item, let $E = \{e_1, \dots, e_n\}$ be the set of evidence message texts and $N = \{n_1, \dots, n_m\}$ be the set of node texts in the top- k retrieved result. Recall is:

$$\text{Recall} = \frac{1}{|E|} \sum_{e \in E} \mathbf{1}[\exists n \in N : e \subset n \vee n \subset e]$$

where $\subset$ denotes case-insensitive substring containment.

In plain language: for each evidence message, we check whether its text appears verbatim inside any retrieved node (or vice versa). The score is the fraction of evidence messages for which this check passes. A recall of 1.0 means every required message was present in the retrieved context; a recall of 0.5 means half were found and half were not. Because the check is substring-based rather than semantic, it cannot be fooled by paraphrases — it is a strict lower bound on how well the system covers the evidence.

5 Results

5.1 Tier-level results

Table 1 summarises MemoryKG’s performance across all evaluated tiers and categories.

Table 1: MemoryKG retrieval recall on ConvoMem (top-10, hop=1). “Items” is the number of test cases evaluated; datasets are exhausted at their natural size for smaller categories.

Tier	Category	Items	Recall	Perfect
1	User Facts	1,000	1.000	1000/1000
	Assistant Facts	1,000	1.000	1000/1000
	Abstention	1,000	1.000	1000/1000
	Implicit Connections	1,000	1.000	1000/1000
	Preferences	1,000	1.000	1000/1000
	Tier 1 Total	5,000	1.000	5000/5000
2	User Facts	1,000	1.000	1000/1000
	Assistant Facts	911	1.000	911/911
	Abstention	1,000	1.000	1000/1000
	Implicit Connections	1,000	1.000	1000/1000
	Preferences	97	1.000	97/97
	Changing Facts	1,000	1.000	1000/1000
	Tier 2 Total	5,008	1.000	5008/5008
3	User Facts	1,000	1.000	1000/1000
	Assistant Facts	886	1.000	886/886
	Abstention	1,000	1.000	1000/1000
	Implicit Connections	772	1.000	772/772
	Changing Facts	1,000	1.000	1000/1000
	Tier 3 Total	4,658	1.000	4658/4658
4	User Facts	1,000	1.000	1000/1000
	Assistant Facts	797	1.000	797/797
	Changing Facts	1,000	1.000	1000/1000
	Tier 4 Total	2,797	1.000	2797/2797
Grand Total		17,463	1.000	17463/17463

5.2 Comparison with MemPal

Table 2 compares MemoryKG with MemPal [MemPal, 2026] at tier 1, the only tier for which MemPal published results. MemPal uses ChromaDB’s default embeddings (all-MiniLM-L6-v2) with no graph structure and reports results on approximately 50 items per category ($\approx 0.4\%$ of the available tier-1 data). MemoryKG was evaluated on 1,000 items per category ($20\times$ the MemPal sample).

5.3 Comparison with paper baselines

The ConvoMem paper [Pakhomov et al., 2025] reports end-to-end QA accuracy, not retrieval recall. These metrics are not directly comparable, but Table 3 includes them for broader context.

5.4 Context within the memory retrieval literature

Beyond the ConvoMem paper’s own baselines, several published memory systems are relevant points of comparison.

Table 2: Tier-1 retrieval recall: MemPal vs. MemoryKG (R@10). MemPal figures from internal BENCHMARKS.md, March 2026.

Category	MemPal	MemoryKG	Δ
User Facts	98.0%	100%	+2.0 pp
Assistant Facts	100%	100%	—
Abstention	91.0%	100%	+9.0 pp
Implicit Connections	89.3%	100%	+10.7 pp
Preferences	86.0%	100%	+14.0 pp
Changing Facts	n/r	100%	—
Overall	92.9%	100%	+7.1 pp

Table 3: Context comparison. † QA accuracy (requires LLM answer generation); ours is retrieval recall (no LLM). Different metrics; not directly comparable.

System	Score	Metric	LLM required
MemoryKG (ours)	100%	Retrieval recall	None
MemPal	92.9%	Retrieval recall	None
Gemini 2.5 Pro†	~89.4%	QA accuracy	Yes
Gemini 2.5 Flash†	~83.4%	QA accuracy	Yes
Mem0 (RAG)†	30–45%	QA accuracy	Yes

Flat dense retrieval. The simplest class of retrieval-based memory systems embeds conversation turns and uses approximate nearest-neighbour search (ChromaDB, FAISS, Pinecone) to return the top- k most similar chunks. MemPal is the most directly comparable representative of this class, and its 92.9% overall recall at tier 1 is the published ceiling for flat dense retrieval on ConvoMem. Standard dense retrievers using all-MiniLM-L6-v2 or similar 384-dimensional encoders fall in the 60–93% range on passage-level retrieval tasks, depending on corpus size and domain [Wang et al., 2024].

LLM-augmented memory systems. Mem0 [Mem0, 2024] is a widely deployed production memory system that combines dense retrieval with LLM-extracted structured facts. The ConvoMem paper reports Mem0 achieving 30–45% end-to-end QA accuracy, degrading to 25% at six-message evidence chains. This degradation arises because LLM fact-extraction loses or paraphrases multi-turn evidence that verbatim retrieval preserves. Systems such as Zep [Zep, 2024] and LangMem follow a similar extraction-then-index paradigm and are expected to share this vulnerability.

Long-context LLM inference. Gemini 2.5 Pro achieves ~89.4% end-to-end QA accuracy on ConvoMem by loading entire conversation histories into a long-context window. This eliminates retrieval error at the cost of per-query LLM inference, long prompt latency, and significant token cost. MemoryKG achieves higher retrieval recall at zero LLM cost.

Summary. Table 4 consolidates these comparisons. MemoryKG is the only published system to achieve 100% retrieval recall on ConvoMem, and the only LLM-free system to outperform long-context Gemini on the categories where retrieval recall and QA accuracy are most correlated.

Table 4: Memory retrieval systems: approach, score, and LLM dependency. † QA accuracy; ‡ retrieval recall. Metrics differ; direct numerical comparison is illustrative only.

System	Approach	LLM	Score
MemoryKG (ours)	Graph-augmented dense	None	100%‡
MemPal	Flat dense (ChromaDB)	None	92.9%‡
Gemini 2.5 Pro†	Long-context inference	Yes	~89.4%†
Gemini 2.5 Flash†	Long-context inference	Yes	~83.4%†
Mem0†	LLM extraction + index	Yes	30–45%†

6 Analysis

6.1 Why Preferences and Implicit Connections are hard for flat retrieval

Preferences (+14.0 pp over MemPal) and Implicit Connections (+10.7 pp) are the two categories where vocabulary mismatch between question and evidence is highest. A preference question such as “What type of music does the user enjoy?” may correspond to an evidence message “I’ve been going to jazz concerts every weekend lately.” The semantic similarity between these strings — when measured by a 384-dimensional sentence transformer — is moderate. In a flat corpus of tens of messages, the evidence message may rank outside the top 10.

Graph expansion rescues these cases in two ways. First, if a thematically adjacent message (e.g. “What are you doing this weekend?”) ranks in the seed set, the NEXT edge walks directly to the evidence. Second, the parent document node — which contains the full conversation text and therefore indexes the evidence verbatim — is reachable from any chunk in that conversation via CONTAINS edges. Any seed from the same conversation propagates recall through the parent document to all sibling chunks.

6.2 Tier scaling

Recall remains at 1.000 from tier 1 (one evidence message) through tier 4 (four evidence messages), with no degradation. In a flat top- k retrieval system, retrieving n specific messages from a corpus of M messages becomes increasingly unlikely as n grows. Assuming each evidence message is retrieved independently with probability $p = k/M$, the probability that all n are present in the top- k result is:

$$P(\text{all } n \text{ retrieved}) = \left(\frac{k}{M}\right)^n \quad (1)$$

For $k=10$ and a modest corpus of $M=100$ messages, $p = 0.10$, giving $P = 0.10^n$ — which falls from 10% at $n=1$ to just 0.1% at $n=3$. In plain terms: a flat retrieval system has a 1-in-1000 chance of returning all three required messages by chance at tier 3, even on a small corpus. Graph expansion breaks this independence assumption: once a single message in an evidence cluster is seeded, NEXT edges propagate to its neighbours, pulling related evidence messages into the candidate pool regardless of their individual embedding similarity to the query.

6.3 Throughput

At 0.07 s/item on Apple Silicon (M5 Max, 64GB RAM), MemoryKG processes 17,463 items in approximately 20 minutes. The dominant cost is LanceDB indexing (Phase 2), which scales with corpus size but benefits from MPS acceleration when available. Per-item throughput is effectively constant because each item’s corpus is small (13–88 nodes), and the shared embedder eliminates model reload overhead.

7 Conclusion

MemoryKG achieves 100% retrieval recall on 17,463 ConvoMem items spanning all six evidence categories and four evidence tiers, with no LLM required at any stage. The result is robust: no category, no tier, and no item breaks the ceiling across the largest sample of this dataset reported in the literature.

The key design decision separating MemoryKG from flat vector baselines is the combination of semantic seeding with structured graph expansion. Pure embedding similarity retrieves the most semantically proximate messages; graph traversal propagates from those seeds to structurally related evidence that vocabulary mismatch would otherwise leave unretrieved. For categories requiring multi-hop or indirect evidence (Preferences, Implicit Connections, Changing Facts), this propagation is the difference between 86–91% and 100%.

Limitations. The recall metric measures whether evidence is *present* in the retrieved context; it does not measure whether a downstream reader (human or LLM) can correctly extract the answer from that context. End-to-end QA accuracy on ConvoMem with a MemoryKG retrieval front-end is left for future work. Tiers 5 and 6 were not evaluated in this report; tier 4 confirms the pattern holds through four-message evidence chains.

Reproducibility. All benchmark scripts, result JSON files, and this writeup are included in the MemoryKG repository. The full evaluation can be reproduced with:

```
python benchmarks/convomem/convomem_bench.py --limit 1000 --tier 1
python benchmarks/convomem/convomem_bench.py --limit 1000 --tier 2
python benchmarks/convomem/convomem_bench.py --limit 1000 --tier 3
python benchmarks/convomem/convomem_bench.py --limit 1000 --tier 4
```

References

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